Load Torque and Permanent Magnet Flux Linkage Estimation of Surface-Mounted PMSM by Using Unscented Kalman Filter

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***Abstract*—In this study, state estimators are designed and compared for surface-mounted permanent magnet synchronous machine (PMSM) drives. The state-space representations used for estimator design are the infinite inertia model, the full model including the equation of motion, and an augmented description extended by the permanent magnet (PM) flux linkage as an additional state variable. By using the latter model, the estimators are less sensitive to the PM flux linkage mismatches due to the online identification of this parameter. For each nonlinear models, observability analyses are presented with and without position measurement. Based on all six models, state estimators are implemented using unscented Kalman filter (UKF) algorithm, and the estimation performances are compared by simulations.**

***Index Terms*—joint state and parameter estimation, nonlinear observability, permanent magnet synchronous machine (PMSM), sensorless estimation, unscented Kalman filter (UKF)**

I. Introduction

In advanced permanent magnet synchronous machine (PMSM) drives, estimators and observers are frequently used to determine the required quantities for the control or monitoring systems without direct measurement. Based on the mathematical description of the drive elements, the faults can be diagnosed as shown in [1], which reviews the state of the art methods for fault diagnostics and fault tolerant control. In addition to diagnosis, estimators can be used to improve the control performance. For example in [2], an estimator-based load torque compensation strategy is proposed for speed controlled high power PMSM drives. Besides load disturbance estimation, the parameters of the drive system can also be estimated and used in adaptive control structure as presented e.g. in [3] and [4]. Work [3] uses recursive least square algorithm for online identification of the electrical parameters of the interior PMSM. These parameters are the temperature dependent stator resistance and the rotor magnetic flux, as well as the machine inductances, which are sensitive to the saturation effect. In contrary, a mechanical parameter – namely the total inertia of rotor and load – is estimated by an extended state observer in [4]. By using the estimated inertia, a feedforward compensation strategy is applied in the speed control loop to achieve better speed response in case of inertia variations. Further significant applications of the estimators are the mechanical sensorless controls. In these drives, the angular position and velocity of the rotor are not measured directly, instead, these variables are estimated based on the measured terminal voltages and currents. In medium and high speed ranges, good performance can be achieved by the fundamental frequency model-based estimators, such as Luenberger-type observer [5], sliding mode observer [6] and nonlinear Kalman filters [7], [8]. However, the efficiency of model-based estimators deteriorates at low speeds, where saliency-based strategies with signal injections are preferred. Unfortunately, signal injection causes torque ripples, increased losses and acoustic noises as mentioned in [9]. Therefore, hybrid solutions are suggested for drives used in a wide speed range. These techniques change the model-based and signal injection-based estimation methods as the speed varies.

A major drawback of mechanical sensorless model-based estimators is the sensitivity to the model parameter mismatches. In other words, the parameter variations may reduce the estimation accuracy, if the parameters are treated as known constants in the estimator, but they differ from the actual values. To reduce the parameter sensitivity, nonlinear Kalman filters are used for joint state and parameter estimation in [10] and [11]. Work [10] presents a speed and position sensorless cubature Kalman filter, which is able to load torque and stator resistance estimation of interior PMSMs. In [10], the current controllers of the drive are adaptively tuned by the estimated resistance value, hence improved control performance is achieved, which is less sensitive for the resistance variations. However, [12] emphasizes that the stator resistance mismatch has less significant effect than the variation of permanent magnet (PM) flux linkage parameter. To reduce the PM flux linkage sensitivity, an adaptive fading extended Kalman filter is proposed to estimate the stator current components, the angular velocity and position of the rotor, the load torque and the flux linkage of PM without using a mechanical sensor in [11]. The proposed estimator is used in model predictive control, which outperforms the conventional control without PM flux linkage estimation.

In this work, state estimators are designed and compared for surface-mounted PMSM drives. For the estimator design, three different state-transition equations are used. The first assumes slowly varying rotor velocity relative to the typical time constants of the electrical quantities. This model is the so-called infinite inertia model as can be seen e.g. in [7]. In contrast, the second state-transition equation includes the equation of motion, and load torque is also an element of the state vector. In the third model, the PM flux linkage is defined as an additional state variable. Thus, the PM flux linkage sensitivity of the estimators can be reduced. During the design process, the observability of all three models is analysed with and without measurement of the angular position. As a result, six state-space models are presented, three with position sensing and three for sensorless applications. The observability conditions for each models are defined by using the locally weak observability theorem introduced in [13]. Due to the nonlinearities, the unscented Kalman filter (UKF) algorithm is used for the implementation of each estimators. Finally, the estimation performance of all six UKFs is compared by simulations.

II. Estimator Design

In this section, the mathematical model of the surface-mounted PMSM is described first. Second, a brief discussion on nonlinear observability can be seen. Then, the state-space models used for estimator design are presented. For each models, observability conditions are defined. Finally, the applied UKF algorithm is presented.

A. Mathematical Model of Surface-Mounted PMSM

To describe the electromagnetic dynamics of the PMSM, the two-axis stationary reference frame model is used in this work. In this coordinate system, the voltage equations are

$$u_\alpha = R i_\alpha + L\frac{\mathrm{d}i_\alpha}{\mathrm{d}t} - \lambda\omega_\mathrm{e}\sin(\varphi_\mathrm{e}) \tag{1}$$

and

$$u_\beta = R i_\beta + L\frac{\mathrm{d}i_\beta}{\mathrm{d}t} + \lambda\omega_\mathrm{e}\cos(\varphi_\mathrm{e}), \tag{2}$$

where u_α, u_β and i_α, i_β are the stator voltage and current components, φ_e and ω_e are the electrical position and velocity of the rotor, λ is the PM flux linkage, as well as R and L are the resistance and inductance of the stator. As can be seen in (1) and (2), the inductance is independent of the rotor direction in case of the surface-mounted type machine, unlike interior PMSMs. Therefore, reluctance torque is not generated in surface-mounted PMSM. Thus, the electromagnetic torque is

$$T_\mathrm{em} = \frac{3}{2}p\lambda\left(i_\beta\cos(\varphi_\mathrm{e}) - i_\alpha\sin(\varphi_\mathrm{e})\right) \tag{3}$$

for three-phase machines. In (3), p denotes the number of pole pairs.

The mechanical behaviour of the machine can be described by the equation of motion:

$$\frac{\mathrm{d}\omega_\mathrm{m}}{\mathrm{d}t} = \frac{T_\mathrm{em} - T_\mathrm{L}}{J}, \tag{4}$$

where ω_m, T_L and J are the mechanical velocity, the load torque and the inertia, respectively. From ω_m, the ω_e electrical velocity can be calculated as

$$\omega_\mathrm{e} = p\omega_\mathrm{m}. \tag{5}$$

B. Brief Description of Locally Weak Observability

For the design of state estimators, it is important to use observable models. In cases of the applied PMSM state-space models, the trigonometric functions and the multiplications of the state variables cause nonlinearities. Therefore, the locally weak observability is used for the analysis of the nonlinear state-space models.

In continuous time, the state-space model of a nonlinear system can be written as

$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = \boldsymbol{f}\left(\boldsymbol{x}, \boldsymbol{u}\right), \tag{6}$$

$$\boldsymbol{y} = \boldsymbol{h}\left(\boldsymbol{x}, \boldsymbol{u}\right). \tag{7}$$

In (6) and (7), $\boldsymbol{f}\left(\boldsymbol{x}, \boldsymbol{u}\right)$ and $\boldsymbol{h}\left(\boldsymbol{x}, \boldsymbol{u}\right)$ are given nonlinear functions of $\boldsymbol{x}$ state and $\boldsymbol{u}$ input vectors, as well as $\boldsymbol{y}$ denotes the output vector. For the system (6)-(7), the observability matrix is

$$\mathcal{O} = \begin{bmatrix} \frac{\partial\mathcal{L}^0_{\boldsymbol{f}}\boldsymbol{h}}{\partial\boldsymbol{x}} & \frac{\partial\mathcal{L}_{\boldsymbol{f}}\boldsymbol{h}}{\partial\boldsymbol{x}} & \dots & \frac{\partial\mathcal{L}^{n-1}_{\boldsymbol{f}}\boldsymbol{h}}{\partial\boldsymbol{x}} \end{bmatrix}^\mathrm{T}, \tag{8}$$

where $\mathcal{L}^k_{\boldsymbol{f}}\boldsymbol{h}$ denotes k^th order Lie derivative of function $\boldsymbol{h}$ with respect to the vector field $\boldsymbol{f}$, and n is the dimension of the state-space. The model (6)-(7) is locally weakly observable at x_0, if the

$$\mathrm{rank}\{\mathcal{O}\}\big|_{x_0} = n \tag{9}$$

rank condition is fulfilled.

C. Infinite Inertia Model

The first state-space model used in this work is the infinite inertia model, which can be given based on (1) and (2). In this model, slowly varying rotor velocity is assumed, therefore $\frac{\mathrm{d}\omega_\mathrm{e}}{\mathrm{d}t} = 0$ can be written instead of the equation of motion. Thus, it is a simplified description, which does not include the mechanical parameters. Since the rotor speed is the first order derivative of the rotor position, the state-transition equation of the infinite inertia model is

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} i_\alpha \\ i_\beta \\ \omega_\mathrm{e} \\ \varphi_\mathrm{e} \end{bmatrix} = \begin{bmatrix} \frac{1}{L}u_\alpha - \frac{R}{L}i_\alpha + \frac{\lambda}{L}\omega_\mathrm{e}\sin(\varphi_\mathrm{e}) \\ \frac{1}{L}u_\beta - \frac{R}{L}i_\beta - \frac{\lambda}{L}\omega_\mathrm{e}\cos(\varphi_\mathrm{e}) \\ 0 \\ \omega_\mathrm{e} \end{bmatrix}. \tag{10}$$

In case of conventional controlled PMSM drives, the angular position of the rotor is also measured in addition to the stator currents. Therefore, in the first case, the output vector contains the i_α, i_β and φ_e variables. Since the model has four state and three output variables, the size of the observability matrix is 12×4, and it can be written as

$$\mathcal{O} = \begin{bmatrix} \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^0 h_3}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_3}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_3}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_3}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_1}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_2}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f h_3}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_3}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_3}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_3}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^2 h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^2 h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^2 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^2 h_1}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^2 h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^2 h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^2 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^2 h_2}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^2 h_3}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^2 h_3}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^2 h_3}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^2 h_3}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^3 h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^3 h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^3 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^3 h_1}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^3 h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^3 h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^3 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^3 h_2}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^3 h_3}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^3 h_3}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^3 h_3}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^3 h_3}{\partial \varphi_\mathrm{e}} \end{bmatrix}. \tag{11}$$

To provide the observability of model (10), when i_α, i_β and φ_e are measured, the observability matrix must have full rank:

$$\operatorname{rank}\{\mathcal{O}\} = 4. \tag{12}$$

To satisfy the rank condition, there must be at least one regular matrix that can be constructed from four different rows of $\mathcal{O}$. In this way, 495 submatrices can be formed. Among these, the following submatrix leads to a significant observability condition:

$$\mathcal{O}_{1-3,6} = \begin{bmatrix} \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^0 h_3}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_3}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_3}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_3}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f h_3}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_3}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_3}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_3}{\partial \varphi_\mathrm{e}} \end{bmatrix}, \tag{13}$$

which is constructed from the 1st, 2nd, 3rd and 6th rows of $\mathcal{O}$. The determinant of $\mathcal{O}_{1-3,6}$ is

$$\det\{\mathcal{O}_{1-3,6}\} = -1. \tag{14}$$

Since, $\det\{\mathcal{O}_{1-3,6}\}$ does not equal to zero, thus $\mathcal{O}_{1-3,6}$ is regular and $\mathcal{O}$ has full rank. Therefore, the model (10) is locally weakly observable at all points of the state-space, when i_α, i_β and φ_e are measured.

For mechanical sensorless applications, only the i_α and i_β currents are measured. In these cases, the size of the observability matrix is therefore only 8×4. The following matrix consists of the first 4 rows of the observability matrix:

$$\mathcal{O}_{1-4} = \begin{bmatrix} \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_1}{\partial \varphi_\mathrm{e}} \\ \frac{\partial \mathcal{L}_f h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_2}{\partial \varphi_\mathrm{e}} \end{bmatrix}. \tag{15}$$

The determinant of $\mathcal{O}_{1-4}$ is

$$\det\{\mathcal{O}_{1-4}\} = \frac{\lambda^2}{L^2}\omega_\mathrm{e}. \tag{16}$$

Since λ and L are positive constants in the infinite inertia model, the position sensorless model is locally weakly observable, when $\omega_\mathrm{e} \neq 0$. This is the same well-known observability condition, which is also presented in [14].

D. Full Model Including the Equation of Motion

The full model includes the equation of motion in addition to the electrical description. Based on (1)-(5), the state-transition equation of the full model can be written as

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} i_\alpha \\ i_\beta \\ \omega_\mathrm{e} \\ \varphi_\mathrm{e} \\ T_\mathrm{L} \end{bmatrix} = \begin{bmatrix} \frac{1}{L}u_\alpha - \frac{R}{L}i_\alpha + \frac{\lambda}{L}\omega_\mathrm{e}\sin(\varphi_\mathrm{e}) \\ \frac{1}{L}u_\beta - \frac{R}{L}i_\beta - \frac{\lambda}{L}\omega_\mathrm{e}\cos(\varphi_\mathrm{e}) \\ \frac{p}{J}\left(\frac{3}{2}p\lambda\left(i_\beta\cos(\varphi_\mathrm{e}) - i_\alpha\sin(\varphi_\mathrm{e})\right) - T_\mathrm{L}\right) \\ \omega_\mathrm{e} \\ 0 \end{bmatrix}. \tag{17}$$

In this model, T_L is defined as an additional state variable, which is assumed to be a slowly varying quantity.

When the rotor position is measured in addition to the stator currents, an important observability condition may be defined based on matrix

$$\mathcal{O}_{1-3,6,9} = \begin{bmatrix} \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f^0 h_3}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_3}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_3}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_3}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_3}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f h_3}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_3}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_3}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_3}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_3}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f^2 h_3}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^2 h_3}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^2 h_3}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^2 h_3}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f^2 h_3}{\partial T_\mathrm{L}} \end{bmatrix}. \tag{18}$$

The determinant of $\mathcal{O}_{1-3,6,9}$ is

$$\det\{\mathcal{O}_{1-3,6,9}\} = \frac{p}{J}. \tag{19}$$

Since p and J are positive constants, $\det\{\mathcal{O}_{1-3,6,9}\}$ does not equal to zero. Therefore, the full model is locally weakly observable at all points of the state-space, if the output vector consists of i_α, i_β and φ_e.

Without position sensing, an important condition can be found for the full model based on matrices

$$\mathcal{O}_{1-5} = \begin{bmatrix} \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_1}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_2}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f^2 h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^2 h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^2 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^2 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f^2 h_1}{\partial T_\mathrm{L}} \end{bmatrix} \tag{20}$$

and

$$\mathcal{O}_{1-4,6} = \begin{bmatrix} \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f h_1}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_1}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_1}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_1}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f h_2}{\partial T_\mathrm{L}} \\ \frac{\partial \mathcal{L}_f^2 h_2}{\partial i_\alpha} & \frac{\partial \mathcal{L}_f^2 h_2}{\partial i_\beta} & \frac{\partial \mathcal{L}_f^2 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial \mathcal{L}_f^2 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial \mathcal{L}_f^2 h_2}{\partial T_\mathrm{L}} \end{bmatrix}. \tag{21}$$

The determinants of these matrices are

$$\det\{\mathcal{O}_{1-5}\} = -\frac{p\lambda^3}{JL^3}\omega_\mathrm{e}\sin(\varphi_\mathrm{e}) \tag{22}$$

and

$$\det\{\mathcal{O}_{1-4,6}\} = \frac{p\lambda^3}{JL^3}\omega_\mathrm{e}\cos(\varphi_\mathrm{e}). \tag{23}$$

The full model without position measurement is locally weakly observable, if $\det\{\mathcal{O}_{1-5}\} \neq 0$ or $\det\{\mathcal{O}_{1-4,6}\} \neq 0$. Since $\frac{p\lambda^3}{JL^3}$ is a positive constant coefficient in (22) and (23), the observability condition is

$$\omega_\mathrm{e} \neq 0, \tag{24}$$

because $\sin(\varphi_\mathrm{e})$ and $\cos(\varphi_\mathrm{e})$ cannot be zero at the same time.

An additional condition may be formulated based on matrices

$$\mathcal{O}_{1,3-6} = \begin{bmatrix} \frac{\partial\mathcal{L}_f^0 h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial T_\mathrm{L}} \\ \frac{\partial\mathcal{L}_f h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_1}{\partial T_\mathrm{L}} \\ \frac{\partial\mathcal{L}_f h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_2}{\partial T_\mathrm{L}} \\ \frac{\partial\mathcal{L}_f^2 h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial T_\mathrm{L}} \\ \frac{\partial\mathcal{L}_f^2 h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial T_\mathrm{L}} \end{bmatrix} \tag{25}$$

and

$$\mathcal{O}_{2-6} = \begin{bmatrix} \frac{\partial\mathcal{L}_f^0 h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial T_\mathrm{L}} \\ \frac{\partial\mathcal{L}_f h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_1}{\partial T_\mathrm{L}} \\ \frac{\partial\mathcal{L}_f h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_2}{\partial T_\mathrm{L}} \\ \frac{\partial\mathcal{L}_f^2 h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial T_\mathrm{L}} \\ \frac{\partial\mathcal{L}_f^2 h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial T_\mathrm{L}} \end{bmatrix} \tag{26}$$

for the full model without position measurement. The determinants of $\mathcal{O}_{1,3-6}$ and $\mathcal{O}_{2-6}$ matrices are

$$\det\{\mathcal{O}_{1,3-6}\} = \frac{Rp\lambda^3}{JL^4}\frac{\mathrm{d}\omega_\mathrm{e}}{\mathrm{d}t}\sin(\varphi_\mathrm{e}) - \frac{2Rp\lambda^3}{JL^4}\omega_\mathrm{e}^2\cos(\varphi_\mathrm{e}) \tag{27}$$

and

$$\det\{\mathcal{O}_{2-6}\} = -\frac{Rp\lambda^3}{JL^4}\frac{\mathrm{d}\omega_\mathrm{e}}{\mathrm{d}t}\cos(\varphi_\mathrm{e}) - \frac{2Rp\lambda^3}{JL^4}\omega_\mathrm{e}^2\sin(\varphi_\mathrm{e}). \tag{28}$$

According to the previously defined condition, the full model is locally weakly observable in position sensorless case, if the angular velocity of the rotor is not zero. However, $\det\{\mathcal{O}_{1,3-6}\}$ or $\det\{\mathcal{O}_{2-6}\}$ is not zero in case of $\omega_\mathrm{e} = 0$, if $\frac{\mathrm{d}\omega_\mathrm{e}}{\mathrm{d}t} \neq 0$ is fulfilled. Hence, this model is locally weakly observable, when ω_e is not zero, or when ω_e varies in standing position. The latter part of the condition is particularly important during speed reversal. It should also be note that this is an extended observability condition, which does not apply to the infinite inertia model.

E. Augmented Model with PM Flux Linkage State Variable

The full model may be augmented as

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} i_\alpha \\ i_\beta \\ \omega_\mathrm{e} \\ \varphi_\mathrm{e} \\ T_\mathrm{L} \\ \lambda \end{bmatrix} = \begin{bmatrix} \frac{1}{L}u_\alpha - \frac{R}{L}i_\alpha + \frac{\lambda}{L}\omega_\mathrm{e}\sin(\varphi_\mathrm{e}) \\ \frac{1}{L}u_\beta - \frac{R}{L}i_\beta - \frac{\lambda}{L}\omega_\mathrm{e}\cos(\varphi_\mathrm{e}) \\ \frac{p}{J}\left(\frac{3}{2}p\lambda\left(i_\beta\cos(\varphi_\mathrm{e}) - i_\alpha\sin(\varphi_\mathrm{e})\right) - T_\mathrm{L}\right) \\ \omega_\mathrm{e} \\ 0 \\ 0 \end{bmatrix}, \tag{29}$$

where λ is also a state variable and not a constant parameter. Although λ is assumed to be a slowly varying quantity.

When the stator currents and the rotor position are also measured, the size of the observability matrix is 18×6. Therefore, 18564 submatrices can be specified to define observability conditions. Among these, the following matrix leads to an important condition:

$$\mathcal{O}_{1,2,4,5,7,8} = \begin{bmatrix} \frac{\partial\mathcal{L}_f^0 h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial \lambda} \\ \frac{\partial\mathcal{L}_f^0 h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial \lambda} \\ \frac{\partial\mathcal{L}_f h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_1}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f h_1}{\partial \lambda} \\ \frac{\partial\mathcal{L}_f h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_2}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f h_2}{\partial \lambda} \\ \frac{\partial\mathcal{L}_f^2 h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial \lambda} \\ \frac{\partial\mathcal{L}_f^2 h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial \lambda} \end{bmatrix}. \tag{30}$$

The determinant of (30) is

$$\det\{\mathcal{O}_{1,2,4,5,7,8}\} = \frac{p}{JL^4}\lambda^3\omega_\mathrm{e}^3. \tag{31}$$

Since $\frac{p}{JL^4}$ is a constant coefficient, while λ and ω_e are state variables, the augmented model is locally weakly observable, when $\lambda \neq 0$ and $\omega_\mathrm{e} \neq 0$ are fulfilled. Note that surface-mounted PMSMs cannot operate without PM flux linkage, so this condition is satisfied in practice if the rotor velocity is not zero.

Without position measurement, the same observability condition may be formulated based on

$$\mathcal{O}_{1-6} = \begin{bmatrix} \frac{\partial\mathcal{L}_f^0 h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f^0 h_1}{\partial \lambda} \\ \frac{\partial\mathcal{L}_f^0 h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f^0 h_2}{\partial \lambda} \\ \frac{\partial\mathcal{L}_f h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_1}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f h_1}{\partial \lambda} \\ \frac{\partial\mathcal{L}_f h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f h_2}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f h_2}{\partial \lambda} \\ \frac{\partial\mathcal{L}_f^2 h_1}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f^2 h_1}{\partial \lambda} \\ \frac{\partial\mathcal{L}_f^2 h_2}{\partial i_\alpha} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial i_\beta} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial \omega_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial \varphi_\mathrm{e}} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial T_\mathrm{L}} & \frac{\partial\mathcal{L}_f^2 h_2}{\partial \lambda} \end{bmatrix}, \tag{32}$$

whose determinant is

$$\det\{\mathcal{O}_{1-6}\} = \frac{p}{JL^4}\lambda^3\omega_\mathrm{e}^3. \tag{33}$$

Note that $\mathcal{O}_{1,2,4,5,7,8}$ and $\mathcal{O}_{1-6}$ are the same matrix, because $\mathcal{O}_{1,2,4,5,7,8}$ does not contain the 3rd rows, which are related to position measurement. Thus, the determinants of these matrices are also the same. Another important note is that the observability conditions formulated for the position sensorless models also provide locally weak observability for the models with position measurement, because the sensorless observability matrices are submatrices of the complete observability matrices by measuring of stator currents and rotor position.

F. UKF algorithm

State estimators are usually implemented in discrete time, because these algorithms are mostly executed on digital signal processors in practice. Therefore, the presented models must be discretized. In this work, the simple Euler method is used for the discrete time approximation of the state-transition equations as

$$\boldsymbol{f}_{\mathrm{d}}\left(\boldsymbol{x}_k, \boldsymbol{u}_k\right) = \boldsymbol{f}\left(\boldsymbol{x}_k, \boldsymbol{u}_k\right) T_{\mathrm{s}} + \boldsymbol{x}_k, \tag{34}$$

where T_{s} is the sampling time, as well as $\boldsymbol{x}_k$ and $\boldsymbol{u}_k$ are the state and input vectors at time k.

Assuming additive process and measurement noises, the discrete time state-space model of a nonlinear system with linear measurement equation can be written as

$$\boldsymbol{x}_{k+1} = \boldsymbol{f}_{\mathrm{d}}(\boldsymbol{x}_k, \boldsymbol{u}_k) + \boldsymbol{w}_k, \tag{35}$$

$$\boldsymbol{y}_k = \boldsymbol{H}\boldsymbol{x}_k + \boldsymbol{v}_k, \tag{36}$$

where $\boldsymbol{y}_k$ is the output vector and $\boldsymbol{H}$ is the measurement matrix. Since the measured variables in the applied PMSM state-space models are also elements of the state vectors, the measurement models are linear. Therefore, a simplified UKF algorithm can be applied similarly to [15]. In (35) and (36), $\boldsymbol{w}_k$ and $\boldsymbol{v}_k$ are independent Gaussian noises for which stochastic hypotheses $\mathbf{E}\{\boldsymbol{w}_k\} = \mathbf{0}$, $\mathbf{E}\{\boldsymbol{v}_k\} = \mathbf{0}$, $\mathbf{E}\{\boldsymbol{w}_k\boldsymbol{w}_k^{\mathrm{T}}\} = \boldsymbol{Q}$ and $\mathbf{E}\{\boldsymbol{v}_k\boldsymbol{v}_k^{\mathrm{T}}\} = \boldsymbol{R}$ hold.

For the prediction of the mean and error covariance of the states, sampling points, also known as sigma points, are used in the UKF. The general unscented transformation applies $2n+1$ sigma points as

$$\boldsymbol{\sigma}_k^{(i)} = \begin{cases} \widehat{\boldsymbol{x}}_k^+, & i = 0, \\ \widehat{\boldsymbol{x}}_k^+ + \left(\sqrt{(n+\kappa)\boldsymbol{P}_k^+}\right)^{(i)}, & i = 1, ..., n, \\ \widehat{\boldsymbol{x}}_k^+ - \left(\sqrt{(n+\kappa)\boldsymbol{P}_k^+}\right)^{(i-n)}, & i = n+1, ..., 2n, \end{cases} \tag{37}$$

where n is the number of state variables, $\widehat{\boldsymbol{x}}_k^+$ and $\boldsymbol{P}_k^+$ are the corrected values of the state vector and the error covariance matrix, as well as κ is a design parameter. In (37), $\left(\sqrt{(n+\kappa)\boldsymbol{P}_k^+}\right)^{(i)}$ is the i^{th} row of matrix $\sqrt{(n+\kappa)\boldsymbol{P}_k^+}$, where the radical symbol denotes Cholesky factorisation. By using function $\boldsymbol{f}_{\mathrm{d}}$, the sigma points can be calculated for the next time step as

$$\tilde{\boldsymbol{\sigma}}_{k+1}^{(i)} = \boldsymbol{f}_{\mathrm{d}}(\boldsymbol{\sigma}_k^{(i)}, \boldsymbol{u}_k), \tag{38}$$

and the state vector and the error covariance matrix can be predicted as

$$\widehat{\boldsymbol{x}}_{k+1}^- = \sum_{i=0}^{2n} \boldsymbol{W}^{(i)} \tilde{\boldsymbol{\sigma}}_{k+1}^{(i)}, \tag{39}$$

$$\boldsymbol{P}_{k+1}^- = \sum_{i=0}^{2n} \left[\boldsymbol{W}^{(i)} \left(\tilde{\boldsymbol{\sigma}}_{k+1}^{(i)} - \widehat{\boldsymbol{x}}_{k+1}^-\right)\left(\tilde{\boldsymbol{\sigma}}_{k+1}^{(i)} - \widehat{\boldsymbol{x}}_{k+1}^-\right)^T\right] + \boldsymbol{Q}, \tag{40}$$

where the weights are

$$\boldsymbol{W}^{(i)} = \begin{cases} \dfrac{\kappa}{n+\kappa}, & i = 0, \\ \dfrac{1}{2(n+\kappa)}, & i = 1, ..., 2n. \end{cases} \tag{41}$$

Due to the linear measurement model, the equations of the linear Kalman filter can be used for the correction of state vector and error covariances as follows:

$$\widehat{\boldsymbol{x}}_{k+1}^+ = \widehat{\boldsymbol{x}}_{k+1}^- + \boldsymbol{K}_{k+1}(\boldsymbol{y}_{k+1} - \boldsymbol{H}\widehat{\boldsymbol{x}}_{k+1}^-), \tag{42}$$

$$\boldsymbol{P}_{k+1}^+ = (\boldsymbol{I} - \boldsymbol{K}_{k+1}\boldsymbol{H})\boldsymbol{P}_{k+1}^-, \tag{43}$$

where $\boldsymbol{I}$ is the identity matrix and the Kalman gain is

$$\boldsymbol{K}_{k+1} = \boldsymbol{P}_{k+1}^- \boldsymbol{H}^T (\boldsymbol{H}\boldsymbol{P}_{k+1}^- \boldsymbol{H}^T + \boldsymbol{R})^{-1}. \tag{44}$$

III. Simulation Results

To compare the performance of the UKF estimators, simulations are carried out under different conditions. The surface-mounted PMSM used for the simulations has 2.8 N·m nominal torque and 419 rad/s nominal mechanical speed. Further parameters of the machine are $p = 4$, $R = 1.9\ \Omega$, $L = 3$ mH, $\lambda = 0.1$ V·s and $J = 0.00018$ kg·m^2. During the simulations, the total load torque is calculated as

$$T_{\mathrm{L}} = D\omega_{\mathrm{m}} + T_{\mathrm{ext}}, \tag{45}$$

where T_{ext} is the external load torque acting on drive system, and the damping coefficient is $D = 0.005$ N·m·s/rad.

For the implementation of the UKF estimators, the following noise parameters are used, which are selected by trial-and-error method. The $\boldsymbol{Q}$ process noise covariance matrices are

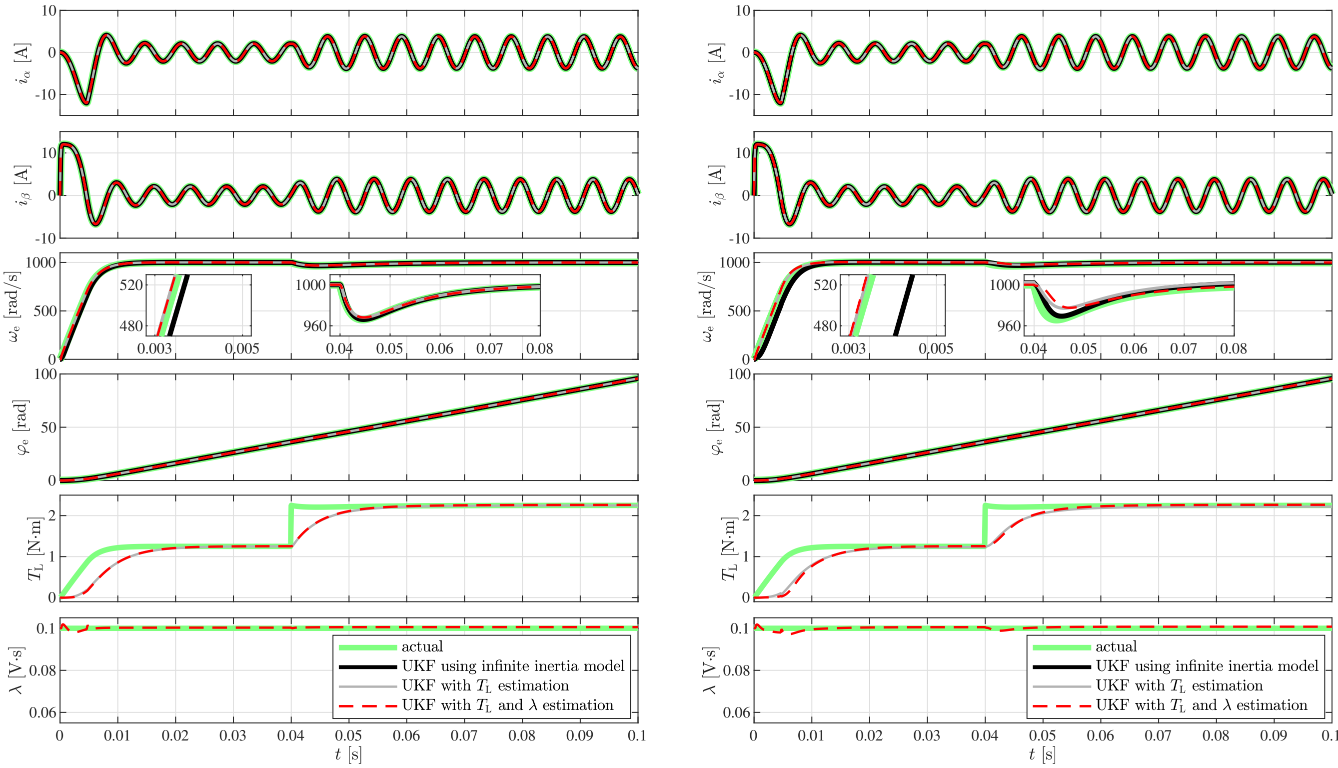
- $\mathrm{diag}\{0.1, 0.1, 100, 10^{-7}\}$ for the infinite inertia model-based UKFs,
- $\mathrm{diag}\{0.1, 0.1, 100, 10^{-7}, 10^{-2}\}$ for the full model-based UKFs,
- $\mathrm{diag}\{0.1, 0.1, 100, 10^{-7}, 10^{-2}, 10^{-5}\}$ for the augmented model-based UKFs,

and the $\boldsymbol{R}$ measurement noise covariance matrices are

- $\mathrm{diag}\{10^{-3}, 10^{-3}, 10^{-5}\}$ for the estimators using current and position sensors,
- $\mathrm{diag}\{10^{-3}, 10^{-3}\}$ for the position sensorless estimators.

The initial value of the error covariance matrix is $\boldsymbol{P}_0^+ = 10^{-4}\boldsymbol{I}$ for all estimators, while the initial states are zeros for i_α, i_β, ω_{e}, φ_{e}, T_{L} state variables, however the nominal 0.1 V·s is used as initial value for λ. The design parameter κ is set to 1 for all state estimators.

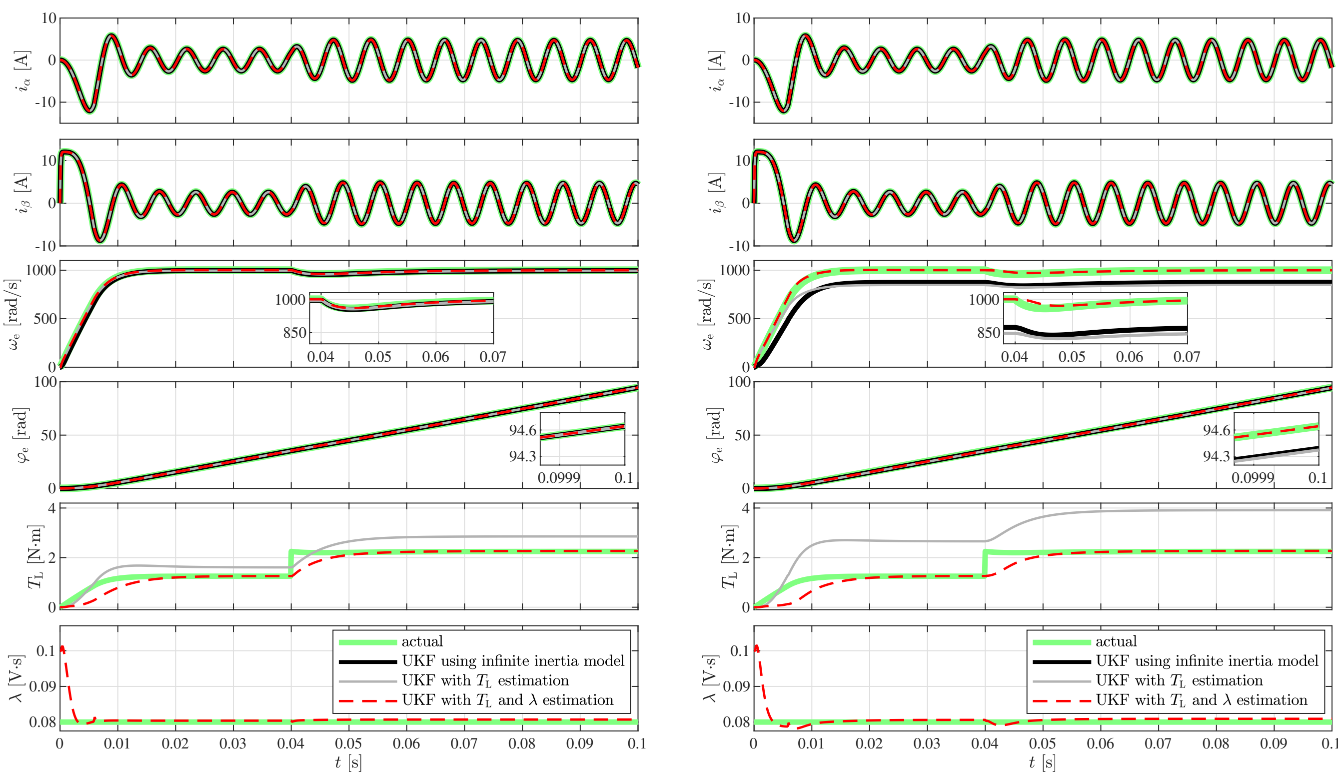
At the beginning of the simulations, the rotor is accelerated from standing position to 1000 rad/s electrical speed without external load torque. At 0.4 s, T_{ext} steps to 1 N·m. In the first simulation, the nominal parameters of the PMSM are used, and the results are shown in Fig. 1. The estimated variables, when



(a) With position measurement.

(b) Without position measurement.

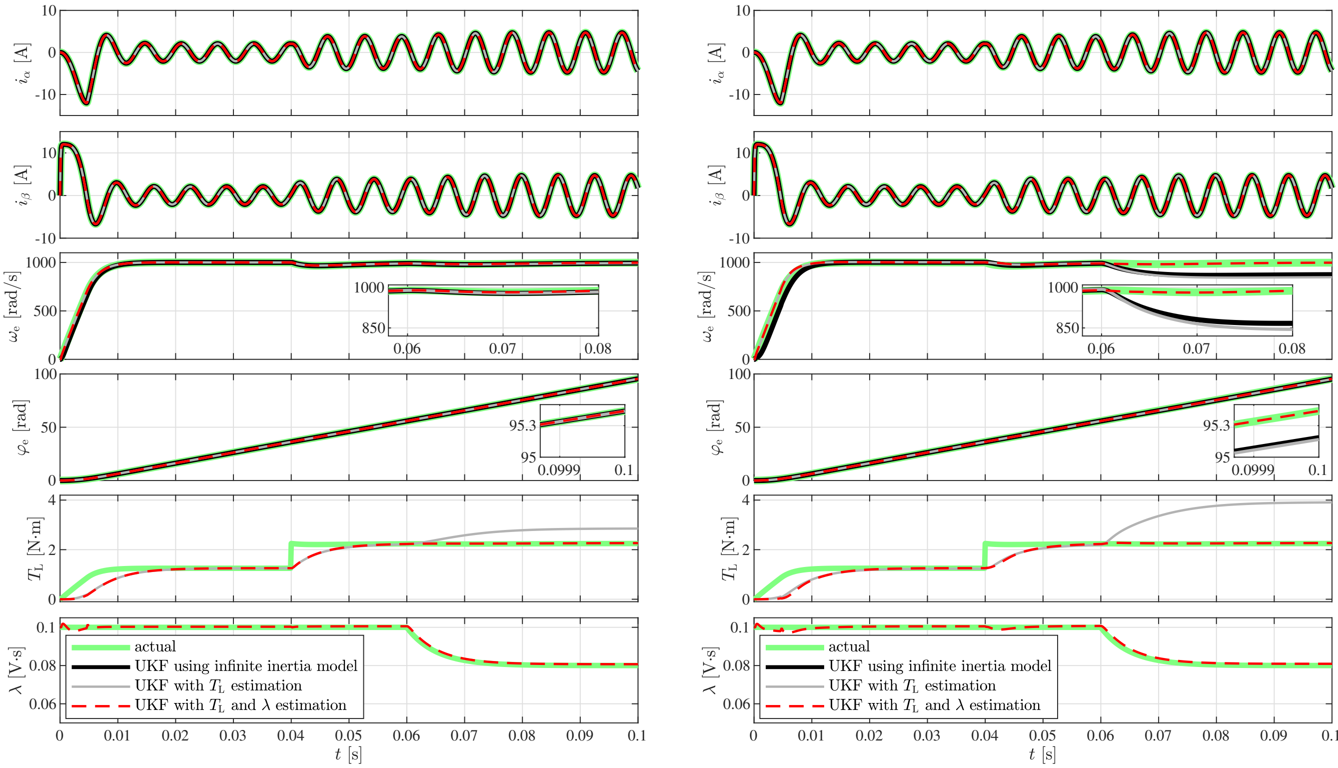
Fig. 1. Performance of the UKF estimators using nominal parameters.



(a) With position measurement.

(b) Without position measurement.

Fig. 2. Performance of the UKF estimators in the case of detuned λ.



(a) With position measurement.

(b) Without position measurement.

Fig. 3. Performance of the UKF estimators besides λ variation.

the rotor position is also measured besides the stator currents, can be seen in Fig. 1(a), while the estimation performances of the position sensorless UKFs are depicted in Fig. 1(b). As the results show, all six state estimators operate with adequate accuracy, when the machine parameters are precisely known. Only the T_L estimation is relative slow during the transients in cases of the full and the augmented model-based UKFs, because slowly varying load torque has been assumed during the design process. However, the estimated values of T_L follow the actual value within 0.01 s. It can also be observed that the position sensorless infinite inertia model-based UKF provides less accurate ω_e estimation during the acceleration compared to the two other state estimators.

The PM flux linkage is also affected by temperature variations and ageing, which can lead to estimation errors in cases of the conventional state estimators. In the second simulation, λ is detuned to 80% of its nominal value in the simulation model of the PMSM. Therefore, λ parameter is inaccurate in the UKFs based on the infinite inertia and full models, while the initial value of λ is also incorrect in the augmented model-based UKF. The results of the simulation with detuned PM flux linkage can be seen in Fig. 2. As Fig. 2(a) shows, i_α, i_β, ω_e and φ_e estimations are accurate in cases of all three estimators, when stator currents and rotor position are also measured. But the load estimation error is increased for the UKF based on the full model. So in this case, using the full model-based estimator in load compensation control can lead to performance deterioration. In contrast, the estimated λ converges quickly to the actual value in case of the augmented model-based UKF, therefore the load estimation is also accurate at steady state. As can be seen in Fig. 2(b), the ω_e and the φ_e estimation performances are reduced for the position sensorless UKFs without λ estimation. In addition, the T_L estimation error is also increased in case of the full model-based UKF. However, the augmented model-based UKF provides sufficient estimation accuracy even without position measurement.

The PM flux linkage can also vary during the operation. For example, increased temperate causes a decrease in λ. In the third simulation, the PM flux linkage starts to decrease at 0.06 s, and then reaches 80% of its nominal value. As the results show in Fig. 3, λ variation causes T_L estimation inaccuracies for the full model-based UKFs. As well as ω_e and φ_e estimation errors can be seen in cases of the UKFs without λ estimation, if the rotor position is not measured. On the other hand, the augmented model-based UKFs with PM flux linkage estimation provide high accuracy both with and without angular position measurement.

IV. Conclusion

In this paper, state estimators have been designed for surface-mounted PMSM using UKF. The first two state-space representations used for the estimators are the conventional infinite inertia model and the full model. The latter also includes

TABLE I
COMPARISON OF ESTIMATION PERFORMANCES FOR THE UKFS USING DIFFERENT STATE-SPACE MODELS

	Infinite inertia model-based UKF		**Full model-based UKF**		**Augmented model-based UKF**	
	with position measurement	**without position measurement**	**with position measurement**	**without position measurement**	**with position measurement**	**without position measurement**
Estimation of the rotor velocity	Accurate	Sensitive to PM flux linkage variations	Accurate	Sensitive to PM flux linkage variations	Accurate	Insensitive to PM flux linkage variations
Estimation of the rotor position	Highly accurate due to the position measurement	Sensitive to PM flux linkage variations	Highly accurate due to the position measurement	Sensitive to PM flux linkage variations	Highly accurate due to the position measurement	Insensitive to PM flux linkage variations
Estimation of the load torque	Not possible	Not possible	Sensitive to PM flux linkage variations	Sensitive to PM flux linkage variations	Insensitive to PM flux linkage variations	Insensitive to PM flux linkage variations
Estimation of the PM flux linkage	Not possible	Not possible	Not possible	Not possible	Accurate and fast under PM flux linkage variations	Accurate and fast under PM flux linkage variations

the equation of motion. In addition, an augmented model was also applied, in which the PM flux linkage is defined as an additional state variable. For each nonlinear models, observability studies were presented both with and without position measurement. To compare the UKF estimators, simulations were carried out. The estimation performances obtained in the simulations are summarised in Table I. These results show the superiority of the augmented model-based UKF over the conventional estimators in the case of PM flux linkage uncertainties.

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